

SHATHURSIMA RAVEENDRAN

Second Year Computer Engineering Undergraduate
University of Peradeniya

Contact no: +94 74 276 4543 Email : shathursimaraveendran04@gmail.com
LinkedIn: www.linkedin.com/in/shathursima-raveendran cn portfolio : thecn.com/SR1960

Profile

Second-year Computer Engineering undergraduate at the University of Peradeniya with a strong interest in Machine Learning, Artificial Intelligence, and Software Development. Passionate about applying engineering and computational techniques to solve real-world problems through innovative and impactful solutions. Actively involved in volunteer initiatives through the IEEE Women in Engineering (WIE) Affinity Group, contributing to the planning and coordination of technical and professional development events. Eager to expand knowledge in machine learning, data-driven technologies, and modern software engineering practices while continuously developing technical, problem-solving, and teamwork skills.

Education

University of Peradeniya Second year Computer Engineering Undergraduate	2024 – 2028
J/Vembadi Girls' High School Secondary Education	2014 – 2023

Experience

Casual Instructor Department of Computer Engineering, University of Peradeniya CO1030 — Data Structures & Algorithms 1 E24 Batch	2026 – Present
Casual Instructor Department of Computer Engineering, University of Peradeniya CO1010 — Programming for Engineers 1 E24 Batch	2025 – 2026
Design Team Member WIE Affinity Group of Student Branch, UOP	2026 – 2027
Editorial Team Member WIE Affinity Group of Student Branch, UOP	2025 – 2026
Editorial Team Member SEDS PERA	2026 – 2027
OGV IR Member AIESEC in Kandy	2025

Projects

Dementia Risk Prediction Model

- Built a dementia likelihood predictor using non-medical factors (demographic, lifestyle, and cognitive data) during a competitive hackathon.
- Implemented Logistic Regression from scratch in NumPy with gradient descent — no ML libraries used for the core model.
- Handled full data pipeline: cleaning, normalization, model serialization (Pickle), and an interactive prediction interface.
- *Skills:* Machine Learning, Python, NumPy, Data Preprocessing

Line Following Robot (EE1010)

- Designed and built an autonomous line-following robot as a semester project in the Electrical & Electronic Engineering module EE1010.
- Programmed an Arduino-based control system using IR sensors to detect and follow a black line path with smooth directional correction.
- Wired and assembled motor driver circuits (L298N), calibrated sensor thresholds, and tuned motor speed differentials for reliable tracking.

- *Skills:* Arduino, Embedded C, Circuit Design, Sensors, Prototyping

Mini Command-Line Shell in ARM32 Assembly (CO1020)

- Developed a functional shell supporting commands: **hello**, **help**, **clear**, **exit**, **len**, and **upper** — entirely in ARM32 Assembly.
- Implemented low-level I/O, memory management, and Linux system calls to simulate a minimal shell environment at the hardware level.
- *Skills:* ARM32 Assembly, Systems Programming, Memory Management

Libtiny3d — 3D Software Rendering Engine

- Built a lightweight 3D rendering engine in C from scratch with no dependency on OpenGL, DirectX, or any graphics API.
- Implemented a full rendering pipeline: custom 3D math library, DDA line drawing, viewport clipping, and Lambertian diffuse lighting.
- *Skills:* C, Makefile, Linear Algebra, 3D Geometry

Signal Intelligence Dashboard — SignalMon

- Built an open-source intelligence system to monitor, process, and visualize real-time socio-economic signals from Sri Lankan news sources.
- Applied NLP pipelines to extract structured strategic insights from unstructured text at scale.
- *Skills:* NLP, Python, Data Engineering, Sentiment Analysis

Volunteering

Coordinator for "CODE WITH WIE2025" · IEEE Women in Engineering

Invigilator & Technical Team Member · ICPC Foundation Contest 2025

Educational Volunteer · WIE Affinity Group of Student Branch, UOP

Volunteer · Yarl IT Hub

Skills

- Strategic Planning
- Problem Solving
- Team Management
- ML/AI
- Software Development
- Basic Networking
- Content Writing
- Canva
- Microsoft Office.

Languages

Tamil (Native) · **English** (Fluent) · **Sinhala** (Basic)

Certifications

- **Get Started with Databricks for Machine Learning** – Simplilearn
- **Google AI Essentials** – Google
- **Java Programming** – Great Learning
- **Introduction to Python** – Sololearn
- **Python for Beginners** – University of Moratuwa